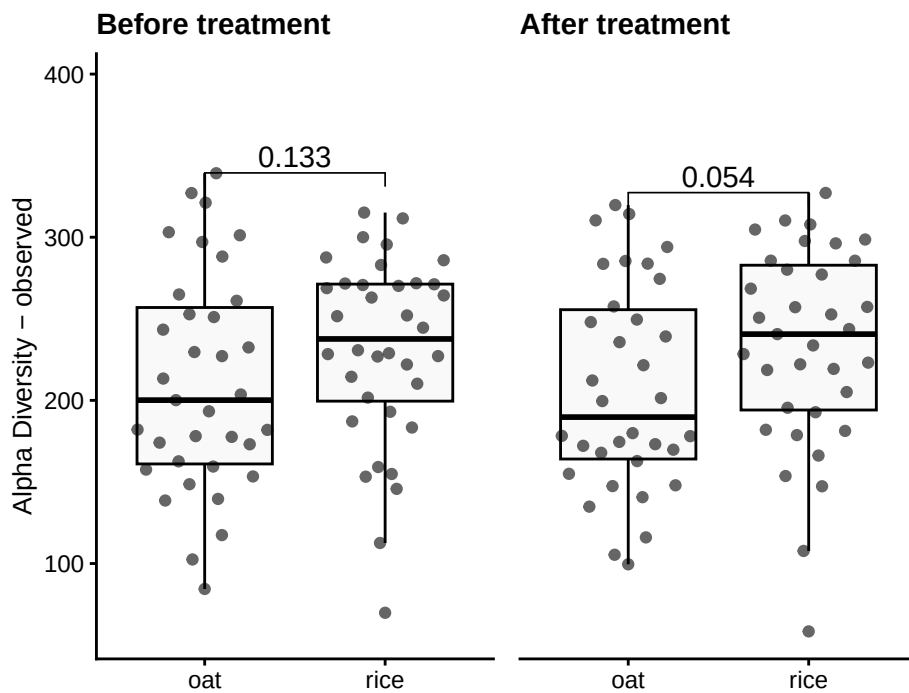


# Alpha Diversity

Muluh

## 1 Group-wise comparisons

- **Index:** observed
- **Group variable:** diet
- **Multiple testing correction:** FDR
- **Statistical test** Wilcoxon



### 1.1 Table of the top findings

#### 1.1.1 Across-group

##### 1.1.1.1 Before treatment

- Multiple testing correction: **FDR**
- Statistical test: **Wilcoxon**

Table 1: Table of alpha diversity comparisons before Treatment

| group1 | group2 | mean1 | mean2 | log2FC | p.adj |
|--------|--------|-------|-------|--------|-------|
| oat    | rice   | 210.9 | 231.3 | 0.13   | 0.133 |

Before treatment, the rice group had a slightly higher observed alpha diversity (231.30) compared to the oat group (210.90). The fold change (log2FC) of 0.13 suggests a small increase in richness for the rice group. However, the corrected p-value ( $p_{adj} = 0.133$ ) indicates that this difference is not statistically significant, meaning it could be due to random variation rather than a true effect of diet.

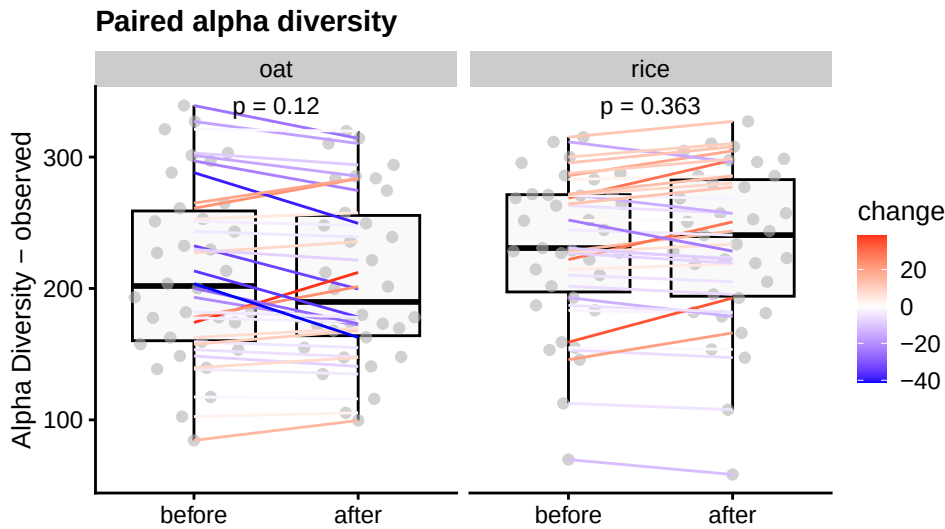
### 1.1.1.2 After treatment

- Multiple testing correction: **FDR**
- Statistical test: **Wilcoxon**

Table 2: Table of alpha diversity comparisons after Treatment

| group1 | group2 | mean1 | mean2 | log2FC | p.adj |
|--------|--------|-------|-------|--------|-------|
| oat    | rice   | 206.9 | 233   | 0.17   | 0.054 |

After treatment, the rice group (233.00) continued to show a higher observed alpha diversity compared to the oat group (206.90), with a fold change (log2FC) of 0.17. The adjusted p-value (p.adj = 0.0540) is approaching statistical significance, suggesting a potential effect of diet on diversity, though it remains marginal.



### 1.1.2 Within-group

#### 1.1.2.1 Paired treatment

- Multiple testing correction: **paired -none**
- Statistical test: **Wilcoxon**

Table 3: Table of alpha diversity paired between timepoint(1 and 2)

| diet | mean1 | mean2 | log2FC | p.value |
|------|-------|-------|--------|---------|
| oat  | 210.9 | 206.9 | 0.04   | 0.120   |
| rice | 231.3 | 233.0 | 0.13   | 0.363   |

Oat group: Observed alpha diversity slightly decreased from 210.90 to 206.90 (log2FC = 0.04, p = 0.120), indicating a small but non-significant reduction over time.

Rice group: Diversity increased modestly from 231.30 to 233.00 (log2FC = 0.13, p = 0.363), but this change is also not statistically significant.

Overall, there were no significant within-group changes in observed alpha diversity over time, suggesting that dietary intervention had a minimal effect on diversity shifts within each group.

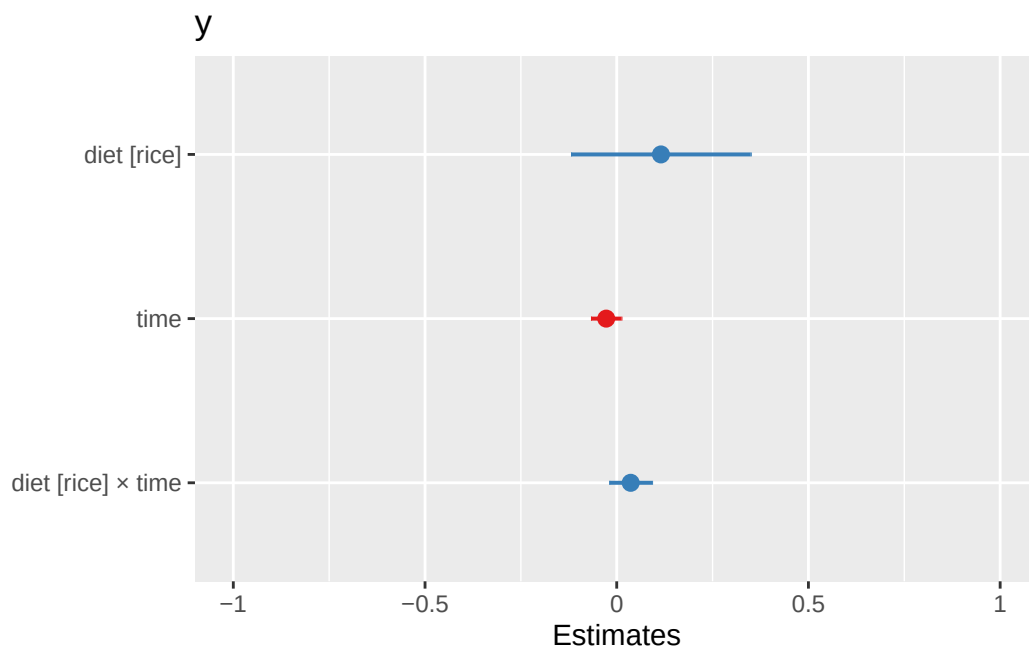
## 1.2 LMM

- Multiple testing correction: **FDR**
- Statistical test: **paired Wilcoxon**

| Characteristic | before |                     |         | after |                     |         |
|----------------|--------|---------------------|---------|-------|---------------------|---------|
|                | Beta   | 95% CI <sup>1</sup> | p-value | Beta  | 95% CI <sup>1</sup> | p-value |
| (Intercept)    | 151    | 67, 235             | <0.001  | 144   | 59, 229             | 0.001   |
| meal_group     |        |                     | 0.4     |       |                     | 0.3     |
| 1              | —      | —                   |         | —     | —                   |         |
| 2              | 32     | -11, 75             | 0.14    | 28    | -16, 72             | 0.2     |
| 3              | 28     | -15, 70             | 0.2     | 30    | -14, 73             | 0.2     |
| 4              | 10     | -32, 53             | 0.6     | -3.8  | -47, 39             | 0.9     |
| age            | 1.0    | -0.44, 2.5          | 0.2     | 1.2   | -0.25, 2.7          | 0.10    |

<sup>1</sup>CI = Confidence Interval

- Formula used: `~ diet * timepoint + (1 | id)`



Diet effect (rice vs. oat): No significant difference between rice and oat groups (**Estimate** = 0.12, **p** = 0.333), suggesting baseline diversity was similar.

Time effect: A slight decrease in diversity over time (**Estimate** = -0.03, **p** = 0.177), but this was not statistically significant.

Diet × Time interaction: The interaction term (**Estimate** = 0.04, **p** = 0.201) suggests that the change in diversity over time did not significantly differ between the oat and rice groups.

## 2 Alternate analyses

- **Beta (Estimate)**: Shows the effect size and direction of a predictor on the outcome. Positive means an increase, negative a decrease.
- **95% CI**: Range where the true beta is likely to fall, reflecting precision.
- **p-Value**: Tests if beta differs from zero;  $p < 0.05$  suggests significance.

## 3 Data Summary

class: TreeSummarizedExperiment

| Paired test between timepoints (1 and 2) | Beta | 95% CI <sup>I</sup> | p-value |
|------------------------------------------|------|---------------------|---------|
| (Intercept)                              | 147  | 84, 210             | <0.001  |
| meal_group                               |      |                     | 0.6     |
| 1                                        | —    | —                   |         |
| 2                                        | 28   | -16, 71             | 0.2     |
| 3                                        | 28   | -15, 71             | 0.2     |
| 4                                        | 2.2  | -27, 32             | 0.9     |
| age                                      | 1.2  | 0.13, 2.2           | 0.026   |

<sup>I</sup>CI = Confidence Interval

```

dim: 1700 140
metadata(0):
assays(1): relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921
rowData names(9): kingdom phylum ... strain clade_name
colnames(140): AE1332 AE2332 ... VK1336 VK2336
colData names(16): sample id ... group observed_log2
reducedDimNames(0):
mainExpName: NULL
altExpNames(20): kingdom phylum ... KO metacyc
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL

```